

Practice Question Set

1. What is an operator?
 2. What is an operand?
 3. How are operators classified based on the number of operands?
 4. What is a unary operator? Give examples.
 5. What is a binary operator? Give examples.
 6. What is a ternary operator?
 7. What are arithmetic operators? List them.
 8. What are relational operators? List them.
 9. What are logical operators? List them.
 10. What are assignment operators? Give examples.
 11. What are bitwise operators?
 12. Explain the purpose of the `sizeof()` operator.
 13. What is the difference between `++` and `--` operators?
 14. What is the purpose of `&` and `*` operators?
 15. Explain the syntax of the ternary operator.
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Important Interview/Viva Questions

16. Difference between unary, binary, and ternary operators.
 17. Difference between relational and logical operators.
 18. Difference between = and ==.
 19. Difference between & and &&.
 20. Difference between | and ||.
 21. Difference between ++i and i++.
 22. Why is % operator called modulus operator?
 23. What is the difference between / and %?
 24. What is bitwise XOR (^)?
 25. What is the use of the ternary operator?
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Answers :

1. What is an operator?

Answer:

An operator is a special symbol that performs mathematical, relational, logical, or bitwise operations on operands.

Example:

a + b

Here, + is the operator.

2. What is an operand?

Answer:

Operands are variables or values on which operators perform operations.

Example:

`a + b`

`a` and `b` are operands.

3. How are operators classified based on the number of operands?

Answer:

1. Unary Operators (1 operand)
 2. Binary Operators (2 operands)
 3. Ternary Operator (3 operands)
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4. What is a unary operator? Give examples.

Answer:

Unary operators work on a single operand.

Examples:

`++`

`--`

`!`

`&`

`*`

`sizeof()`

5. What is a binary operator? Give examples.

Answer:

Binary operators require two operands.

Examples:

+

-

*

/

%

>

<

&&

||

=

6. What is a ternary operator?

Answer:

The ternary operator is the only operator requiring three operands.

Syntax:

```
(condition) ? expression1 : expression2;
```

Example:

```
(a>b) ? a : b;
```

7. What are arithmetic operators?

Answer:

Arithmetic operators perform mathematical calculations.

Operator	Operation
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus

8. What are relational operators?

Answer:

Relational operators compare two values and return True (1) or False (0).

Operators:

>
<
>=
<=
==
!=

9. What are logical operators?

Answer:

Logical operators combine conditions.

Operators:

&&
||
!

10. What are assignment operators?

Answer:

Assignment operators assign or update values.

Examples:

=
+=
-=
*=
/=

%=

Example:

```
a += 5;
```

means

```
a = a + 5;
```

11. What are bitwise operators?

Answer:

Bitwise operators manipulate individual bits.

Operators:

&

|

^

~

<<

>>

12. What is the purpose of `sizeof()`?

Answer:

`sizeof()` returns the memory occupied by a variable or datatype in bytes.

Example:

```
sizeof(int);
```

Usually returns 4 bytes.

13. What is the difference between `++` and `--`?

Answer:

- `++` increases a value by 1.
- `--` decreases a value by 1.

Example:

```
a++;  
a--;
```

14. What is the purpose of & and * operators?

Answer:

- & returns the address of a variable.
- * accesses the value stored at that address.

Example:

```
int x=10;  
int *ptr=&x;
```

15. Explain the syntax of the ternary operator.

Answer:

```
(condition)?expression1:expression2;
```

If condition is true, expression1 executes; otherwise expression2 executes.

Example:

```
max= (a>b) ?a:b;
```

Important Interview Answers

16. Difference between Unary, Binary, and Ternary Operators.

Unary	Binary	Ternary
One operand	Two operands	Three operands
++, --	+, -, *, /	?:
Works on one value	Works on two values	Conditional operator

17. Difference between Relational and Logical Operators.

Relational	Logical
Compare values	Combine conditions

Relational	Logical
>, <, ==	&&
Return 1 or 0	Return 1 or 0

18. Difference between = and ==.

=	==
Assignment operator	Equality operator
Assigns value	Compares values

Example:

```
a = 10;
a == b
```

19. Difference between & and &&.

&	&&
Bitwise AND	Logical AND
Works on bits	Works on conditions
Evaluates both operands	Short-circuit evaluation

20. Difference between | and ||.

Bitwise OR	Logical OR
Operates on bits	Operates on conditions

21. Difference between ++i and i++.

Answer:

Pre-increment

```
++i
```

First increments, then uses the value.

Post-increment

`i++`

First uses the value, then increments it.

Example:

```
int i=5;
```

```
x=++i;    // x=6
```

```
y=i++;    // y=6, i becomes 7
```

22. Why is % called the modulus operator?

Answer:

Because it returns the remainder after division.

Example:

```
10 % 3
```

Output:

```
1
```

23. Difference between / and %.

/	%
Quotient	Remainder
$20/4 = 5$	$20\%3 = 2$

24. What is bitwise XOR (^)?

Answer:

XOR returns 1 when bits are different and 0 when they are same.

Example:

```
1 ^ 0 = 1
```

```
1 ^ 1 = 0
```

```
0 ^ 0 = 0
```

25. What is the use of the ternary operator?

Answer:

The ternary operator is a shorthand form of `if-else`.

Example:

```
max = (a > b) ? a : b;
```

It selects one of two values based on a condition.